Uncooperative 0.5 m satellite in LEO. Approach from 1 km to 1 m, no attitude or navigation signal, rotation up to 2 deg/s. Speed at most 0.1 m/s, and 0.01 m/s below 5 m. Pointing +/-2 deg. GNC needs a vision update every 200 ms. The centroid is the centre of the box.

## 1. Methodology

- Angular size theta = 0.5 / range. Pixels on target = theta / IFOV, with IFOV = pixel size / focal length.
- Detector: YOLOv8s, one class, 640 input. 480 train / 120 validation, split by box size. 100 test images held out.
- Loss: CIoU for the box, classification BCE with label smoothing, distribution focal loss. AdamW, dropout 0.2.
- Dark and blurred test images are matched in training by brightness, blur and CLAHE. Test time only resizes to 640.
- Tracker: corners inside the box, Lucas-Kanade flow, detector correction every 15 frames. Bearing from camera.json.

## What the camera has to measure

| Range | Angular size | Requirement |
|---|---|---|
| 1 km | 0.029 deg | 3 pixels needs IFOV <= 0.17 mrad/px |
| 100 m | 0.29 deg | Line of sight, then a rough range |
| 50 m | 0.57 deg | 50 pixels for pose: IFOV <= 0.20 mrad/px |
| 10 m | 2.9 deg | Field of view about 9 deg |
| 1 m | 29 deg | Field of view about 35 deg, or the target is clipped |

In one 200 ms cycle the chaser moves 2 cm at 0.1 m/s, and the target can rotate 0.4 deg.

## Camera A against Camera B

| | Camera A | Camera B |
|---|---|---|
| IFOV and field of view | 0.078 mrad/px, 20 deg | 0.69 mrad/px, 70 deg |
| Power and frame rate | 3.2 W, 18 fps | 1.8 W, 60 fps |
| Pixels at 1 km / 100 m | 6.4 / 64 | 0.7 / 7 |
| Pixels at 15 m / 1 m | 440 / clipped | 50 / 730 |
| Power, 10 fps, close FOV | Fail, pass, fail | Pass, pass, pass |

Selection: Camera B. It meets the 3 W limit, 10 fps, and still contains the target at 1 m. It cannot detect at 1 km (0.7 pixel). Acquisition beyond about 250 m needs a prior or a second camera. Camera A sees the target at 1 km, but it is over the power cap and its 20 deg lens clips a 29 deg target at 1 m.

### Lighting

A glint pulls the box centre off the body.

### Earth and stars

A brightness detector locks onto clutter.

### Scale and rotation

The target grows about 1000x and tumbles. One fixed template loses it.

**On-board chain**

Camera B → YOLOv8s detect → Flow tracker → Line of sight → GNC at 5 Hz

The navigation filter and GNC are specified, not coded. Flow runs every frame. The detector only starts the box or repairs it.

## Interfaces and timing

| From | What is passed | Rate | Budget |
|---|---|---|---|
| Camera | Frame and its time | 5 fps, 10 fps under 10 m | 50 ms |
| Detector | One box and a score | Start, then every 15 frames | 100 ms |
| Tracker | Centre, bearing, coast flag | Every frame | 20 ms |
| Filter to GNC | Bearing and rough range | 5 Hz | 20 ms |

About 170 ms of that budget sits inside the 200 ms cycle. A bearing is usable from about 240 m, pose from about 15 m. At 1 km this chain must not claim a detection.

## What was already run, and what is still a ground test

| Check | Result |
|---|---|
| Validation, 120 images | 94.2% detection at IoU 0.5, median centre error 13.3 px |
| Test file, 100 images | One box each in test_predictions.json. Labels are hidden. |
| Tracking, 300 frames | 280 flow updates, 19 detector corrections, angle 6.8 to 1.9 deg |

### Simulation

250 m to 1 m, with sun, shadow, 2 deg/s rotation and Earth in view.

### Hardware-in-the-loop

Camera B on a known model. Time the chain and count dropouts.

### Flight-like cases

Eclipse, glint, Earth behind the target, lost track inside 10 m.

## 2. Results - validation (120 labelled images)

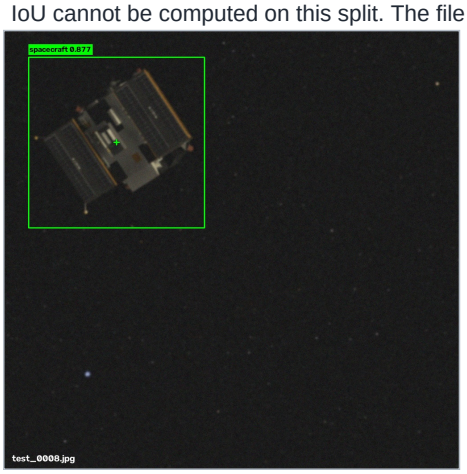
| | | | |
|---|---|---|---|
| **94.2%**<br>Detection rate<br>IoU >= 0.5, 113/120 | **0.76**<br>Median IoU<br>mean 0.73 | **13.3 px**<br>Median centroid error<br>mean 29 px | **0.975**<br>mAP50<br>mAP50-95 is 0.55 |

Qualitative. On a good detection the box sits on the spacecraft and 13 px is a small centre error on a 1024 image. The mean error is worse than the median because a missed box is counted as a large offset.

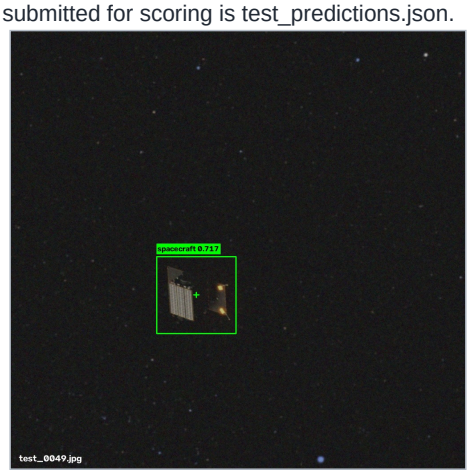
## Test split (100 images, labels hidden)

| | | | |
|---|---|---|---|
| **100/100**<br>Images with a box<br>test_predictions.json | **0.72**<br>Median score<br>mean 0.65, max 0.88 | **83**<br>Scores at or above 0.50 | **8**<br>Scores below 0.25<br>ids 82, 94, 99 weakest |

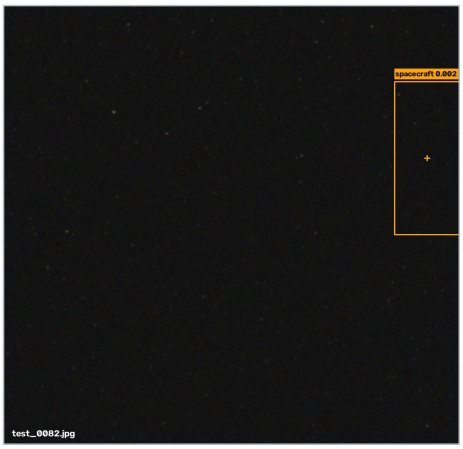
IoU cannot be computed on this split. The file submitted for scoring is test_predictions.json.




test_0008, score 0.88. Clear target, tight box.

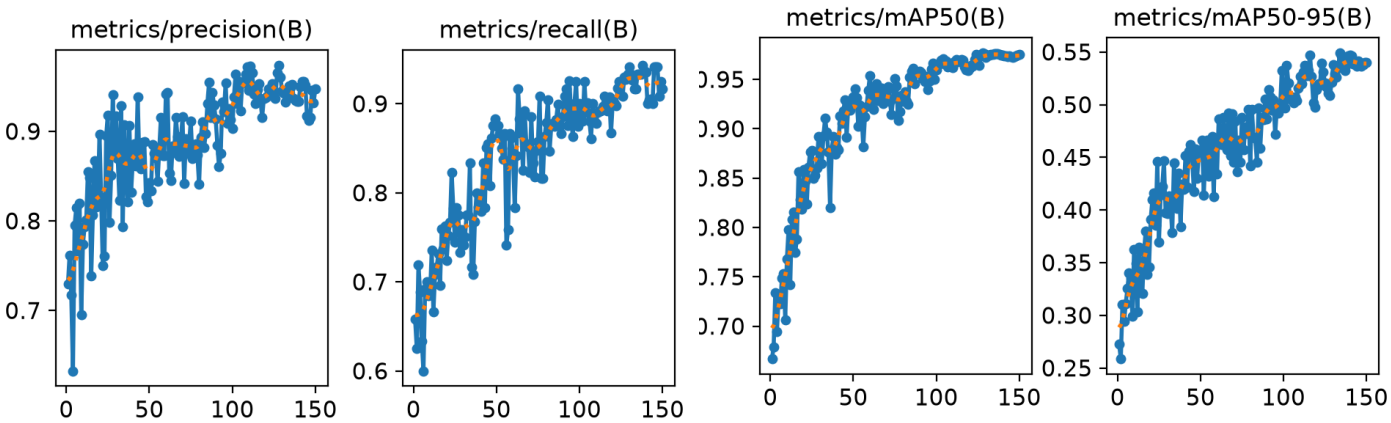



test_0049, score 0.72. Smaller target, still enclosed.




test_0082, score 0.002. Box on an empty edge. A miss.

The eight low scores are the ones a hidden label is likely to reject. High-score frames put the cross on the body.

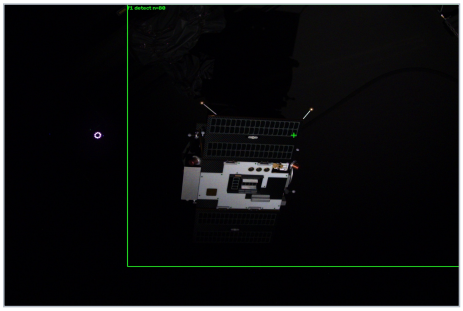


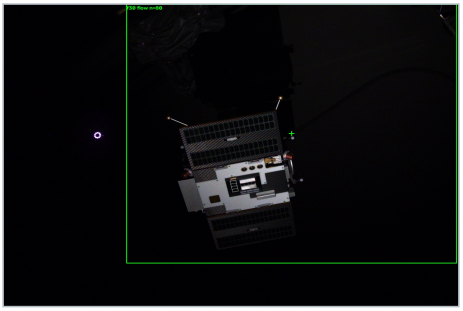

*Validation curves over 150 epochs. mAP50 levels off near 0.98. mAP50-95 near 0.55: boxes are usable, not pixel-tight.*

## Tracking (300 frames, no ground-truth boxes)

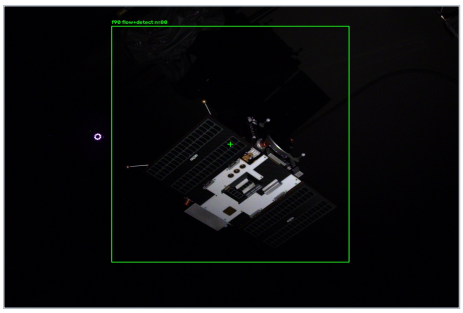
| 280 | 19 | 0 | 6.8 to 1.9 deg |
|---|---|---|---|
| Frames by optical flow | Detector corrections plus frame 1 | Frames where flow failed | Angle off the boresight |

Centroid moves from (916, 468) to (683, 583). Path length 1210 px. The detector resets the box every 15 frames, so the centre steps rather than gliding. The frames below keep the box on the spacecraft while it tumbles.

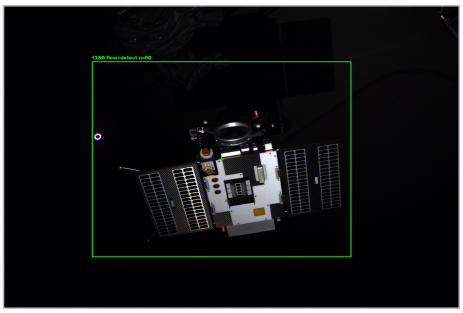

Frame 1. Detector starts the box.

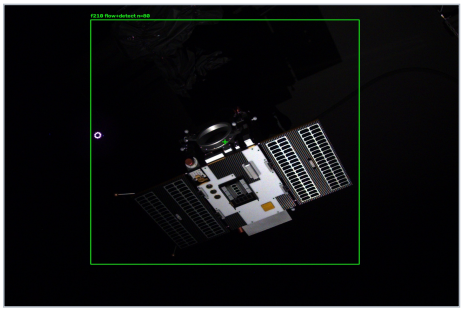

Frame 30. Flow carries the box.

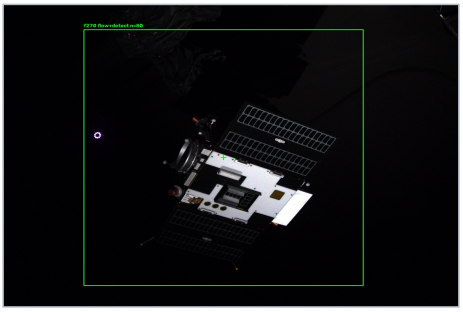

Frame 60. Still on the body.

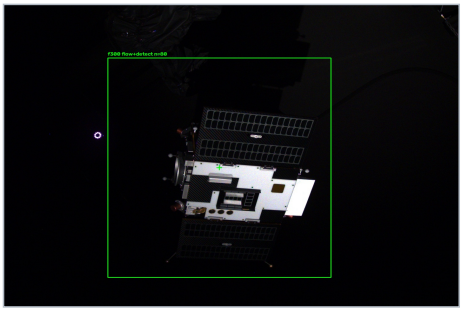

Frame 90. Target is tumbling.


Frame 150. Box follows the new outline.


Frame 180. Features stay on the body.


Frame 210. Detector has resized the box.


Frame 270. Track still locked.


Frame 300. Held to the last frame.

## 3. Limitations

- Camera B is under one pixel at 1 km, so it cannot do acquisition at that range.
- Range from the box, R ~ 0.5/theta, is only a proxy until the target is about 50 pixels (about 15 m on Camera B).
- The test set is darker and more blurred. There is no denoise step at test time, only the training augmentation and a resize.
- The sequence has no labels, so tracking has no IoU. The filter and GNC are not implemented. Inside 10 m, do not follow a stale bearing for more than one or two cycles.

## 4. Conclusions

- Fly Camera B under the written power and field-of-view limits. Do not expect it to acquire the target at 1 km.
- On labelled validation data the detector is usable: 94.2% detection at IoU 0.5, median IoU 0.76, median centre error 13.3 px.
- On the unlabelled test set every image has a box, 83 of 100 scores are at least 0.5, and 8 scores are weak.
- The tracker holds the target for all 300 frames (280 flow updates, 19 detector corrections) and moves the line of sight from 6.8 deg to 1.9 deg off the boresight.